CECH COHOMOLOGY OF AN INJECTIVE ABELIAN SHEAF IS CONCENTRATED IN DEGREE ZERO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all degrees.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Cech cohomology of a covering with coefficients in an abelian sheaf. Same chapter, the Cech complex of a covering and the description of its cohomology by maps from the representable sheaves attached to the fibre products. Supplies the groups computed in the statement.
- Injectivity of an abelian sheaf and the extension property. Same chapter, the characterisation of injective abelian sheaves by extendability of maps along injections. Supplies the vanishing argument for the positive degrees.

Source reference. Original lemma 03F6: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1319–1319. Navigation pointer invoked at proof line 1352; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03AU: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1189–1189. Navigation pointer invoked at proof line 1355; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03AN: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 851–851. Navigation pointer invoked at proof line

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1357; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{U} = \{U_i \to U\}_{i \in I}$ be a covering of $\mathcal{C}$. Let $\mathcal{I}$ be an injective abelian sheaf, i.e., an injective object of $Ab(\mathcal{C})$. Then*

$$\check{H}^p(\mathcal{U}, \mathcal{I}) = \begin{cases} \mathcal{I}(U) & \text{if} \quad p = 0 \\ 0 & \text{if} \quad p > 0 \end{cases}$$

Proof. By Lemma 03F6 we see that $\mathcal{I}$ is an injective object in $PAb(\mathcal{C})$. Hence we can apply Lemma 03AU (or its proof) to see the vanishing of higher Čech cohomology group. For the zeroth see Lemma 03AN. □